

Yangmengfei (Murphy) Xu

PhD Candidate in Robotics

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Education

PhD	The University of Melbourne , Engineering and IT - Expected thesis submission: late 2026. - Research Focus: Control systems and sensing technology in autonomous navigation for mobile robots in complex environments. - Rowden White Scholarship (awarded to the highest-ranked applicant in the department; one recipient per year). - Melbourne Research Scholarship.	Melbourne, Australia 2022 – present
MPhil	The University of Melbourne , Engineering and IT - Thesis: "Inducing human movement pattern change." - Melbourne Research Scholarship (Fee Offset). - Developed a novel robotic framework to induce human motor adaptation through manipulation of motor cost, without explicit error feedback. - Conducted human subject experiments to validate the effectiveness of the proposed approach, demonstrating significant improvements in motor learning and adaptation. - Score: 90/100	Melbourne, Australia 2019 – 2022
MEng	The University of Melbourne , Mechatronics Engineering	Melbourne, Australia 2016 – 2018
BEng	The University of Liverpool , Electrical & Electronic Engineering	Liverpool, UK 2012 – 2016
BEng	Xi'an Jiaotong-Liverpool University , Electrical & Electronic Engineering	Suzhou, China 2012 – 2016

Research Experience

Control and Planning Design for Autonomous Robots in Mining Engineering

Researcher

- Developed robust control and planning methods for autonomous and multi-robot systems, including Spatial Iterative Learning Control, Active Disturbance Rejection Control for non-holonomic robots, and geometry-based distributed connectivity maintenance for visual-sensing agents.
- Demonstrated disturbance rejection in harsh environments (Unmanned Systems, 2023), connectivity preservation under uncertainty (Unmanned Systems, 2024), and spatial learning control under input saturation and variable task duration (IEEE/ASME Transactions on Mechatronics, 2026).

Human Motor Adaptation in Rehabilitation Robotics

Researcher

- Designed and conducted human experiments to induce implicit motor adaptation through energy-based cost shaping using a manipulandum platform; developed the Indirect Shaping Control framework for movement retraining.
- Demonstrated indirect and implicit motor adaptation through robot-generated force fields (IEEE Transactions on Neural Systems and Rehabilitation Engineering, 2021), and established a quantitative link between human motor cost and movement adaptation (IFAC, 2026).

Autonomous Community-Sharing Service Robot

Project Leader

- Designed and developed an autonomous community-sharing robot to foster social connection and sustainability through item exchange and collective memory creation.
- Demonstrated how robotics can support community belonging and environmental responsibility; the prototype was presented at the HRI 2025 Student Design Competition and was awarded the HRI 2025 Sustainability Recognition.

Epistemic Logic Modeling of Human Theory of Mind

Project Leader

- Developed the Predictive Justified Perspective (PJP) model for representing and predicting higher-order beliefs in dynamic multi-agent environments, with applications to human-robot interaction and epistemic planning.
- Investigated bounded human reasoning through pilot user studies (CHI EA, 2025) and predictive belief updates for dynamic epistemic planning using PJP (SoCS, 2026).

PyPose Open-Source Robotics Library

Developer

- Contributed core capabilities to PyPose, an open-source robotics library with 1.6k+ GitHub stars, focusing on point-cloud processing and registration; co-authored the PyPose v0.6 paper presented at the IROS 2023 Workshop.

Experience

The University of Melbourne & Boton Technology, Research Engineer

Melbourne, Australia

- Led an industry university collaboration developing autonomous cleaning robots for material handling systems in the mining sector.
- Responsible for system design, autonomous navigation, computer vision, and AI decision-making modules for large-scale industrial deployment.

2021 – 2023

2 years

The University of Melbourne, Course Tutor and Project Supervisor

Melbourne, Australia

- Deliver tutorials and laboratory workshops across engineering and robotics subjects; prepare teaching materials and assessments, mark assignments and examinations, and provide structured feedback to students.
- Co-supervised more than 30 capstone and research projects, providing guidance on project planning, experimental design, implementation, and weekly progress, with selected projects leading to publishable research outcomes.

2021 – present

5 years

AI Cubed Academy, Robotics Instructor

Melbourne, Australia

- Delivered robotics and programming courses to K-12 students, fostering interest in STEM fields.
- Developed curriculum and hands-on projects to enhance student engagement and learning outcomes.

2020 – 2022

2 years

Volunteer

Ausdroid AI and Robotics Student Group, Founder and President

Melbourne, Australia

- Grew the community to 300 members, organizing regular meetups, workshops, and guest lectures from academic and industry professionals.
- Established partnerships with local and global tech companies to provide internship and project opportunities for members.

2017 – present

Suzhou Ocean Paradise Autistic Children's Rehabilitation Center, Volunteer

Suzhou, China

Nov 2012 – Nov 2013

Suzhou Full House Charity Shop, Committee Member

Suzhou, China

Sept 2012 – Sept 2015

Awards

Engineering and IT Conference Travel Scholarship The University of Melbourne	Oct 2025
Sustainability Recognition Award 2025 20th ACM/IEEE International Conference on Human-Robot Interaction (HRI)	Mar 2025
Rowden White Scholarship The University of Melbourne	Dec 2022
Melbourne Research Scholarship The University of Melbourne	Sept 2022
ICRA RoboMaster AI Challenge Third Prize IEEE International Conference on Robotics and Automation (ICRA)	2021, 2019
Melbourne Research Scholarship The University of Melbourne	July 2019
ICRA RoboMaster AI Challenge Finalist IEEE International Conference on Robotics and Automation (ICRA)	2018

Publications

Spatial Iterative Learning Control for Nonholonomic Robotics With Input Saturation Early access. Xu, Yangmengfei, Tan, Ying, Oetomo, Denny doi.org/10.1109/TMECH.2026.3721852 (IEEE/ASME Transactions on Mechatronics)	2026
Motor Cost Re-Optimization in Indirect Human Movement Pattern Adaptation Presented in Busan, South Korea. Xu, Yangmengfei, Crocher, Vincent, Fong, Justin, Tan, Ying, Oetomo, Denny (23rd IFAC World Congress)	Aug 2026
Beyond Static Assumptions: the Predictive Justified Perspective Model for Episodic Planning Volume 19, issue 1, pages 65–73. Hu, Guang, Li, Weijia, Xu, Yangmengfei doi.org/10.1609/socs.v19i1.43074 (Proceedings of the International Symposium on Combinatorial Search)	Aug 2026
Modeling Higher-order Human Beliefs Using the Justified Perspective Model Li, Wanchun, Zhang, Chenyuan, Li, Weijia, Hu, Guang, Xu, Yangmengfei doi.org/10.1145/3706599.3720223 (ACM CHI EA 25)	Apr 2025
From Isolation to Connection: Community Service Robots for Social Cohesion and Sustainability Xu, Yangmengfei, Tian, Suxuan, Wang, Guojing, Tang, Bin doi.org/10.1109/HRI61500.2025.10973824 (ACM/IEEE HRI 2025)	Mar 2025
A Geometry-Based Distributed Connectivity Maintenance Algorithm for Discrete-time Multi-Agent Systems with Visual Sensing Constraints Li, Xiaoli, Fu, Jinyun, Liu, Mingliang, Xu, Yangmengfei, Tan, Ying, Xin, Yangbin, Pu, Ye, Oetomo, Denny doi.org/10.1142/S2301385024410097 (Unmanned Systems)	2024
On Active Disturbance Rejection Control for Unmanned Tracked Ground Vehicles with Nonsmooth Disturbances Liu, Mingliang, Xu, Yangmengfei, Lin, Xuteng, Tan, Ying, Pu, Ye, Li, Wen, Oetomo, Denny doi.org/10.1142/S2301385024500353 (Unmanned Systems)	2024

PyPose v0.6: The Imperative Programming Interface for Robotics

2023

Zitong Zhan, Xiangfu Li, Qihang Li, Haonan He, Abhinav Pandey, Haitao Xiao, Yangmengfei Xu, Xiangyu Chen, Kuan Xu, Kun Cao, Zhipeng Zhao, Zihan Wang, Huan Xu, Zihang Fang, Yutian Chen, Wentao Wang, Xu Fang, Yi Du, Tianhao Wu, Xiao Lin, Yuheng Qiu, Fan Yang, Jingnan Shi, Shaoshu Su, Yiren Lu, Taimeng Fu, Karthik Dantu, Jiajun Wu, Lihua Xie, Marco Hutter, Luca Carlone, Sebastian Scherer, Daning Huang, Yaoyu Hu, Junyi Geng, Chen Wang

arxiv.org/abs/2309.13035 (IEEE/RSJ IROS 2023 Workshop)

Inducing Human Motor Adaptation Without Explicit Error Feedback: A Motor Cost Approach

2021

Xu, Yangmengfei, Crocher, Vincent, Fong, Justin, Tan, Ying, Oetomo, Denny

doi.org/10.1109/TNSRE.2021.3096516 (IEEE Transactions on Neural Systems and Rehabilitation Engineering)

Magnetic-based Soft Tactile Sensors with Deformable Continuous Force Transfer Medium for Resolving Contact Locations in Robotic Grasping and Manipulation

2019

Mohammadi, Alireza, Xu, Yangmengfei, Tan, Ying, Choong, Peter, Oetomo, Denny

www.mdpi.com/1424-8220/19/22/4925 (Sensors)

Skills

Programming: Python, C++, C, MATLAB/Simulink, LabVIEW

Tools and Frameworks: ROS/ROS2, PyTorch, CUDA, OpenCV, Docker, Git

CAD Software: SolidWorks

PCB Design: Altium Designer, JLC EDA

Languages: English, Chinese